Choupo

The Open Source Chemical Process Simulator

EduTools Guide

Choupo-dev

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1 What this guide is — and what it is not

EduTools is the plane of Choupo's browser application where the *classical graphical methods* are constructed over curves the engine computes: the McCabe–Thiele staircase, the Kremser recovery curve, the ε–NTU chart, the Levenspiel plot, the Van Heerden ignition diagram, the pinch composite curves, and the rest. Each tool draws a construction a student has met on paper, on numbers a real solver produced, and puts the method's answer beside the engine's.

> **Key idea.** This guide is a **lesson companion**, not a theory manual. It says what each construction is *for*, what to turn, what to watch, and what the picture is silent about. It contains no derivations at all.

That absence is deliberate and it is architectural. Every equation behind these tools already has exactly one home — `docs/theoryGuide.pdf` — and every live tool declares, in its registry entry, the named destination in that guide where its derivation lives. A second document carrying the same derivations would be a second home for the same statement, and within one semester the two would disagree. So this guide *points*, and the Theory Guide *derives*. When you want to know why $\varepsilon = (1-e)/(1-C_r e)$, follow the pointer. When you want to know what a class will get wrong about it, you are in the right document.

The five questions each tool section answers

Every tool documented here is written to the same five headings, in the same order, because a lesson is more portable than a description:

1. **The question** — in one sentence a student would recognise from a problem sheet.
2. **What to vary, what to watch** — the specific knob that makes the lesson visible, and what should happen when you turn it.
3. **The misconception the construction exposes** — what students actually get wrong, not merely what is true. This is the part worth reading twice.
4. **What the picture does not say** — the limits of the construction. In this project that is treated as part of teaching it, never as an apology for it.
5. **Where the derivation lives** — a pointer into the Theory Guide, never a copy of it.

Where the other material is

- **Theory Guide** — every derivation, every hypothesis, every worked example. The authority for all physics.
- **Explorer Guide** — the *property surfaces* (T-x-y, $\gamma(x)$, P^{sat} scans). The split criterion between the two planes is stated in the application itself: method-construction → EduTools, property-surface → Explorer.
- **Tutorials Guide** — the runnable cases. Every EduTools tool runs one of them; this guide names which.
- **User Guide** — authoring cases on disk, which is where any finding from a tool should end up.

2 How a tool feeds itself

> *What you'll learn.* Where the numbers on the screen come from, why a tool works with no case open, and what a knob is actually allowed to change.

2.1 Opening a tool

EduTools is a tab of its own. The tool chooser leads the chrome row on that tab; on a case tab showing EduTools in its body the chooser is one slim row above the construction. Either way the list is the same, and a construction that is on the roadmap but not yet built is shown *disabled* rather than hidden — the menu states the roadmap instead of implying it.

Every tool is deep-linkable:

```
?workspace=methods&tool=<id>
```

The `methods` key is a contract and does not change with the workspace's caption, so a link already in a student's notes goes on working. The tool identifiers are the ones listed in §3 and §6.

Under the chooser sits a one-line caption of what the construction teaches, and beside it a book icon that opens the Theory Guide *at this tool's own section.* That icon is the pointer this whole guide is built around: a tool that cannot name its derivation is not finished, and a test in the application refuses an anchor the Theory Guide does not define.

2.2 Classroom, or your own run

Most tools default to **Classroom**: the tool clones a bundled tutorial — its *witness* case — writes your knob values into that case's own dictionary files, and solves it in the browser with the WebAssembly build of the same solver the command line runs. Nothing needs to be open, and nothing on disk is touched.

When the application's current run also carries the surfaces a tool reads, a source toggle offers **Current run** beside the classroom witness. A tool that a run cannot feed says so and stays empty; it never draws something else wearing the same name.

> **Key idea.** A knob turns a *declared scalar* in the witness's dictionary and nothing else. The engine then recomputes everything. A knob that cannot find its key fails loudly rather than silently doing nothing, and a knob that would not reach the physics is deliberately absent — each absence recorded with its reason in the tool's own source.

That last rule has teeth. Several plausible-looking controls are missing on purpose: a pressure that a liquid-phase rate law never reads would be a placebo control; an integrator step count sitting beside physical knobs invites reading numerics as physics; a kinetic pre-exponential that a witness deliberately re-tuned would leave the tool teaching a case other than the one it names.

2.3 Why the picture can be trusted

Three standing rules govern this plane, and they are worth telling a class about, because they are the reason the construction and the simulator cannot quietly diverge.

1. **No physics in the browser.** Every curve is an engine column, an engine KPI, or an engine-written report file. The browser draws.
2. **Method geometry is the one exception, and it is labelled.** A staircase, a trapezoid area, a textbook closed form, a heat cascade re-run over published curves — these *are* the method being taught, so a few tools own them. Where that happens the tool says so on screen, and the engine's own number is the judge: agreement is verification, and a disagreement is a finding about the drawing, never a moved answer.
3. **An absence stays an absence.** A number the engine did not publish is not rendered as zero and not invented. Solver advisories are lifted out of the run log and shown, because these tools have no log tab to bury them in.

> **Common pitfall.** A tool that shows nothing is usually telling you something. "This run publishes no Van Heerden surface" means the reactor you solved was isothermal — there is no generation-versus-removal diagram for it. The honest empty state is a diagnosis; read it before changing the tool.

3 The constructions

Six constructions are documented in full below. The remaining six live tools are listed in §6 with what is missing from each, because a thin lesson is worse than an admitted gap.

The six were chosen to span the three transfer subjects a course actually has to cover, two constructions each, and to prefer the tools whose *misconception* is both common and hard to dislodge by telling:

- **Mass transfer** — McCabe–Thiele (§3.1) and Kremser (§3.2).
- **Heat transfer** — ε–NTU (§3.3) and pinch composite curves (§3.4).
- **Reaction** — Levenspiel (§3.5) and Van Heerden (§3.6).

3.1 Distillation: the McCabe–Thiele staircase

Declared anchor: `ch:distillation`.

The question

A binary feed of composition x_F is to be split into a distillate x_D and a bottoms x_W. At reflux ratio R and feed quality q, how many equilibrium stages does it take, and on which stage does the feed enter?

What to vary, what to watch

> **Try it now.** Set a sharp split ($x_D = 0.95$, $x_W = 0.05$, $x_F = 0.5$), then drag R *down* toward the red $R_{\min}$ tick on the slider. Watch three things at once: the staircase steps multiply, the *approach to pinch* bar fills and reddens, and at $R \le R_{\min}$ the construction stops with a banner naming the pinch abscissa. Then drag R up and watch how little the stage count changes once you are clear of the cliff.

- R — the primary knob. The slider can be bound to R directly or to $R/R_{\min}$; the second binding is the design lens ("1.3× minimum") and makes the cliff a fixed landmark at 1.0 instead of a number that moves.
- q — rotates the q-line about (x_F, x_F). Watch the feed-stage index move, and watch $R_{\min}$ itself change: the same column, the same specifications, a different minimum reflux, because the feed arrived hotter or colder.
- E_{MV} — the Murphree tray efficiency. Below 1 the staircase steps to a dashed *pseudo*-curve and the count becomes a count of real trays.
- **The partial-condenser switch** — adds exactly one equilibrium stage, and the readout says so.
- **The components, γ model and P** — these are the only settings that re-run the engine, because they are the only ones that change the equilibrium curve. Everything else redraws instantly.

The misconception the construction exposes

> **Common pitfall. That the equilibrium curve responds to the reflux ratio.** It is the commonest confusion in the whole subject, and it is usually invisible because on paper the student redraws the entire figure each time. Here the curve is a frozen engine result and the staircase is pure geometry over it, so turning R at sixty frames a second while $y^*(x)$ does not so much as flicker makes the point physically: *reflux is an operating decision; the equilibrium curve is a statement about the mixture.* Change the pressure or the activity model and the engine re-runs and the curve moves — which is the same lesson from the other side.

Two further confusions the same picture settles:

- **That $R_{\min}$ is a property of the column.** It is the intersection of the q-line with the equilibrium curve — feed composition, feed quality, product specifications and thermodynamics, and no equipment at all. Turn q and $R_{\min}$ moves although nothing has been built or unbuilt.
- **That more reflux is better because it needs fewer stages.** The chart shows only one half of that trade. Stage count falls steeply just above $R_{\min}$ and then flattens; the vapour traffic $V = (R+1)D$, and with it the reboiler duty, the condenser duty and the column diameter, goes on rising linearly and is *nowhere on this diagram.* A student who optimises on the staircase alone will choose a column that cannot be paid for.

What the picture does not say

- **The operating lines are straight because Constant Molar Overflow is assumed.** The tool states this permanently under the chart rather than in a footnote. CMO is an energy assumption (equimolar latent heats),

and it is the one thing on the diagram that is not the real model. The equilibrium curve, by contrast, is the real curve at that pressure — not a constant-α toy.
- **No energy, no hydraulics, no diameter.** There is no reboiler duty, no condenser duty, no pressure drop, no flooding check. A column that is impossible to build converges perfectly well here.
- **Stages are not trays** unless $E_{MV} < 1$, and E_{MV} is a *declared* number — from a correlation, a chart, or plant data. The engine will not invent it.
- **The azeotrope you see belongs to the declared model.** If no curated binary-interaction pair covers the system, the tool announces that it is drawing *ideal* mixing and that the diagram therefore cannot show an azeotrope at all. Read that banner: an absent azeotrope may be a fact about the mixture or a fact about the data coverage, and only the banner tells you which.
- **Binary only.** A third component has no axis here.

Where the derivation lives

Derivation. Theory Guide, *Distillation column, Wang-Henke bubble-point* (`ch:distillation`) — the stage model, CMO, and the rigorous column this construction is the hand method for. The closed forms for the two limits the staircase locates geometrically, N_{min} and R_{min}, are derived in *Shortcut distillation: Fenske–Underwood–Gilliland* (`ch:fug`). The Murphree efficiency and its effective-K treatment are in *Tray hydraulics* (`ch:hydraulics`). The staircase construction itself — horizontal to the operating line, vertical to the equilibrium curve — is drawn and explained in the absorber chapter (`ch:absorber`), where the same geometry serves a counter-current cascade.

3.2 Absorption: the Kremser shortcut, judged

Declared anchor: `ch:absorber`.

The question

A gas carrying a soluble solute is washed counter-currently with a lean solvent. How much of the solute do N stages recover — and does the one-line shortcut agree with what the column actually does?

What to vary, what to watch

The construction has two halves and the lesson is in the gap between them. The curve is the Kremser closed form, recovery against N at the absorption factor $A = L/(KV)$ the engine published. The point is the engine's own stagewise recovery for the column that was solved. The chip between them is the labelled deviation.

> **Try it now.** Start from the witness (an ammonia/water absorber, six stages, both feeds at 298.15 K). Now raise the ammonia in the gas feed, step by step, and watch *two* chips together: `dT_rise` climbing as more heat of absorption is released, and the deviation between the shortcut and the column growing with it. Then put the ammonia back and raise the solvent flow instead: A rises, recovery rises, and the deviation shrinks.

- **Stages** N — the knob students reach for first. Watch how quickly the curve flattens: past a handful of stages, more stages buy very little.
- **Solvent flow** L — the knob that actually moves the answer, because it moves A.
- **Solute feed flow** — the knob that breaks the shortcut, by making the column non-isothermal.
- **Feed temperatures** — both of them, gas and solvent, with the `nonIsothermal` flag and `dT_rise` as the readout.

The misconception the construction exposes

> **Common pitfall. That the deviation is an error.** A student who has only met the closed form reads any gap between shortcut and simulator as a defect — a bad K, a loose tolerance, a bug. It is neither. Kremser assumes *one constant* A, which means a straight operating line and a straight equilibrium line, which means an isothermal column. Absorption is exothermic. The solvent heats as it descends, K rises with temperature, and $A = L/(KV)$ therefore *falls* exactly where the column is doing its work. The engine re-evaluates K stage by stage; the shortcut cannot. The gap is the assumption failing, and it is real.

The direction matters as much as the size, and it is the part worth labouring: for a soluble, exothermic solute the isothermal Kremser is *optimistic*. It promises recovery the column will not deliver. A shortcut that erred the other way would merely be conservative; this one sends you to site with a column that underperforms.

A second confusion, milder but persistent:

- **That $A > 1$ settles the matter.** A is a ratio of two slopes, not a pass mark. At A barely above one, recovery approaches its ceiling so slowly that no realistic stage count gets there; at A comfortably above one, a few stages do almost everything. The curve makes the difference between those two situations obvious in a way that the inequality $A > 1$ never can.

What the picture does not say

- **The A on the chip is a single number for a column whose K varies.** It is evaluated at the feed temperature. That is precisely why the two sides of the deviation chip are not the same calculation — and precisely what makes the chip informative.
- **Absorbers only.** A stripper publishes a stripping factor $S = KV/L$ and a temperature *drop*, not these keys, so it does not activate this tool. That is a correct refusal, not a missing feature: the mirrored construction is a different lesson.
- **Stages are theoretical.** No packing height, no HETP, no tray efficiency, no flooding.
- **No solvent circuit.** The lean solvent arrives lean. A real plant regenerates it, and the regenerator's residual solute sets a floor on the recovery that no stage count can beat. That loop is not on this diagram.

Where the derivation lives

Derivation. Theory Guide, *Absorber and stripper: the counter-current cascade* (`ch:absorber`). The absorption factor $A_i = L/(K_i V)$ and the tridiagonal stage system are in *The counter-current cascade and the absorption factor*; the reason the engine does not use the closed form, and the temperature bulge that makes the isothermal shortcut optimistic, are in *Why the energy balance matters.*

3.3 Heat exchange: effectiveness against NTU

Declared anchor: `ch:hx-entu`.

The question

I have an exchanger of area A and overall coefficient U, and two streams with known inlet temperatures and flows. What duty do I actually get — and if I want more, will more area give it to me?

What to vary, what to watch

The chart is the family of ε(NTU) curves at several capacity ratios C_r, with the solved exchanger's own point placed on the curve for its own C_r. A chip reports whether the drawn geometry reproduces the engine's effectiveness; when it does not, that is a statement about the configuration the tool assumed, never about the point.

> **Try it now.** Double the area, from 10 to 20 m^2, and read the duty before and after. Then double it again, to 40. Note how much of the first doubling you got back on the second. Now set the two heat-capacity rates far apart (raise one flow, drop the other), so $C_r \to 0$, and switch between counter-current and co-current: the two curves close up until they lie on top of each other.

- **Area A and coefficient U** — both enter only through $\text{NTU} = UA/C_{\min}$, which is the first thing to notice: a better coefficient and a bigger exchanger are the same move on this chart.
- **The two flows** — they set $C_{\min}$, $C_{\max}$ and hence both NTU and C_r, so a flow change slides the point along one curve *and* moves it to another.
- **The two inlet temperatures** — they do not move the point at all. They set $Q_{\max}$, and therefore the duty in kilowatts, but ε, NTU and C_r are all independent of them. Verifying that is a good five minutes.
- **The configuration** — counter-current, co-current, or one shell with two tube passes.

The misconception the construction exposes

> **Common pitfall. That duty scales with area.** NTU is linear in A; ε is emphatically not. Every curve on the chart bends over, and past roughly NTU = 3 the return on further area is nearly nothing — the exchanger is already delivering most of what the inlet temperatures permit, and no amount of metal buys what thermodynamics has capped. Students who have only seen $Q = UA\,\Delta T_{LM}$ read a proportionality that is not there, because ΔT_{LM} collapses as the exchanger gets good.

The second one is more interesting, and the chart proves it in one move:

- **That counter-current is always better.** It is never worse, but at $C_r = 0$ — one stream condensing or boiling, or simply so large that its temperature barely moves — the two arrangements give *identical* effectiveness, and the curves coincide exactly. The advantage of counter-current flow is purchased entirely by the temperature change of the smaller stream, so when that change is absent there is nothing to buy. This is worth doing on the screen because it is the kind of fact that is remembered as a slogan and applied where it does not hold.
- **That ε is an efficiency, so $\varepsilon = 0.75$ means a quarter is wasted.** Nothing is wasted; the exchanger is adiabatic to its surroundings and every joule that leaves the hot stream arrives in the cold one. ε is normalised by $Q_{\max} = C_{\min}(T_{h,\mathrm{in}} - T_{c,\mathrm{in}})$, a ceiling set by the *inlet temperatures*, which is a statement about what an infinitely large exchanger could do — not about losses.

What the picture does not say

- **It is a rating result, not a design.** "Given this hardware, what happens?" is the question answered. "What hardware do I need?" is a different loop, with pressure drop and geometry in it.
- **No pressure drop, no fouling, no cost.** The chart's flat tail says extra area stops paying in duty; it says nothing about what the area costs, and nothing about the pumping power the extra length consumes.
- **Heat-capacity rates are taken as constant** across the pass. A stream whose c_p swings, or which changes phase part way, is outside the assumption.
- **Three configurations only.** Cross-flow arrangements, and the correction factors that go with them, are not offered here.

Where the derivation lives

Derivation. Theory Guide, *Two-stream heat exchanger: the ε–NTU rating method* (`ch:hx-entu`) — the definitions of C_r and NTU, the counter-current and co-current closed forms, the duty from $Q_{\max}$, and the $UA\,\mathrm{LMTD} = Q$ self-check. The design problem the rating chart does not answer — geometry, pressure drop, the sizing loop — is the section immediately following it, *Heat exchanger DESIGN*.

3.4 Heat integration: the composite curves and the pinch

Declared anchor: `ch:pinch`.

The question

Before a single exchanger is drawn: what is the least hot utility and the least cold utility this set of streams can possibly need?

What to vary, what to watch

The tool draws the engine's own published composite curves, reports its targets $Q_{H,\min}$ and $Q_{C,\min}$ and the pinch temperature, re-runs the heat cascade over the published curves as a cross-check, and renders the engine's candidate-match table verbatim.

> **Try it now.** Change nothing but $\Delta T_{\min}$ — from 20 K down to 10, then up to 30. Watch the composite curves slide together and apart, the two targets move in opposite directions, and the pinch temperature itself change. Nothing physical has been altered: not a stream, not a flow, not a temperature. Ask the class what, then, has changed.

- $\Delta T_{\min}$ — the decisive knob, and the one that is a *choice*.
- **The supply temperatures** — move one stream's start and watch whether the pinch stays where it was. Sometimes it does not, and the region boundaries in the candidate table move with it.
- **The flows** — a flow scales that stream's CP, which changes the slope of its segment of the composite curve. A steep segment and a flat one are the same duty over different temperature spans.
- **The candidate table** — read the temperature-feasibility column and the pinch-rule column separately. A match may be thermodynamically possible and still violate the CP rule at the pinch.

The misconception the construction exposes

> **Common pitfall.** **That the pinch is a property of the plant.** It is a property of the stream data *plus a number the engineer chose.* Turning $\Delta T_{\min}$ moves the pinch temperature and both targets while the process stands still. Students routinely quote "the pinch is at 115°C" as though it had been measured; it was assumed into existence, half of it by the choice of driving force, and the value of that choice is an economic trade-off between exchanger area and utility bill that this pass does not make for you.

Two more, and the third is the one that decides coursework marks:

- **That a target is a design.** The table lists *thermodynamically admissible candidates* and never ranks them. The word "optimal" does not appear anywhere in this pass, and a gate in the test suite enforces that. Reaching the target may require more exchangers, and more awkwardly placed ones, than anybody would build; plot plan, control, start-up and safety are not in the temperature table.
- **That heating below the pinch merely costs a little extra.** It costs *exactly its own duty*, and so does cooling above the pinch, and the two are additive:

$$\sum(\text{heating below}) + \sum(\text{cooling above}) \;=\; Q_{H,\text{current}} - Q_{H,\text{min}}.$$

  The violation diagnostics report both sums, and they are computed by different code from the targets — so the identity closing is a check you can do by hand and the tool can fail. A student who has only the slogan "do not transfer heat across the pinch" cannot say how much a violation costs; this identity says exactly how much.

What the picture does not say

- **Energy targets only.** Area targets, capital targets, and the capital-versus-energy trade-off that would let you *choose* $\Delta T_{\min}$ economically are a further phase and are not implemented. The Theory Guide states this in the same words rather than describing something the engine does not have.
- **It never rewrites the network.** The flowsheet you wrote is the flowsheet that ran; the pass analyses it and stops.
- **A unit enters as its net duty.** A column with a reboiler and a condenser is a single net number unless you model the two as separate units — and for a column that is usually the wrong picture. Splitting is the fix, and it is your decision, not the pass's.
- **The grand composite curve is derived in view.** The engine publishes the hot, cold and shifted composite curves; the tool builds the grand composite from those, and labels it as such. The cross-check chip exists so you can see that the derivation reproduces the engine's targets.

Where the derivation lives

Derivation. Theory Guide, *Pinch analysis: the target before the network* (`ch:pinch`) — the CP line, why $\Delta T_{\min}$ shifts the axes, the problem table cascade by cascade, the composite curves, the candidate matches, the violation identity, and an explicit *What this pass does not do.*

3.5 Reactor sizing: the Levenspiel plot

Declared anchor: `ch:pfr`.

The question

The same reaction, the same feed, the same conversion — which reactor is smaller, a plug-flow tube or a stirred tank, and by how much?

What to vary, what to watch

One chart carries $1/(-r_A)$ against conversion X_A, and on it two areas: the plug-flow reactor's integral $\int_0^{X_{\text{out}}} \mathrm{d}X/(-r_A)$ under the curve, and the stirred tank's rectangle of height $1/(-r_A)$ *at the outlet* and width X_{out}. Both witnesses are twins — byte-identical feed, kinetics and thermodynamics — so the comparison is between reactors and never between chemistries.

> **Try it now.** Raise the plug-flow reactor's volume so X_{out} climbs, and watch the rectangle and the area separate. At low conversion they nearly coincide; at high conversion the rectangle towers over the area. Then lower the temperature: the whole curve lifts (a slower rate is a larger ordinate) and both reactors grow —

but the *ratio* between them is governed by the curve's shape, not its height.

- **Reactor temperature** — through the Arrhenius constant.
- **The two feed flows** — the limiting reactant sets F_{A0} and the co-reactant changes what the rate law sees.
- **The plug-flow volume** — the knob that walks X_{out} along the curve, which is the axis the whole lesson lives on.

The misconception the construction exposes

Common pitfall. That the stirred tank needs more volume because it is somehow a worse reactor. Students reach for a hardware explanation — poorer mixing, dead zones, short-circuiting. The chart shows that the cause is the opposite and is purely geometric: a stirred tank is *perfectly* mixed, so every molecule in it sits at the outlet composition and therefore at the outlet rate — the *lowest* rate anywhere in the duty. The tube sees the whole range of rates from inlet to outlet. The rectangle circumscribes the area because perfect mixing throws away every fast rate the reaction was willing to give. *Being well mixed is what costs the volume.*

And immediately after that, the guard against over-learning it:

- **That "a CSTR always needs more volume" is a law.** It follows from the curve *rising* with conversion, which is what positive-order kinetics do. Draw a rate law whose $1/(-r_A)$ has a minimum — autocatalysis, strong product inhibition — and the rectangle can be the smaller of the two over part of the range, which is why real plants sometimes put a tank first and a tube after it. The construction is valuable exactly because it is a picture and not a rule: it shows you how to check, on any kinetics, rather than what to remember.
- **That the smaller reactor is the cheaper one.** Volume is not price. A large stirred tank can be considerably cheaper than a small tubular reactor with the same duty once you count fabrication, catalyst containment, cleaning and control.

What the picture does not say

- **The comparison is isothermal.** Both witnesses run at the same declared temperature. Nothing here shows the thermal behaviour that makes the two reactors genuinely different animals — for that, go to §3.6.
- **No residence-time distribution.** Plug flow and perfect mixing are the two ideal limits; every real reactor is between them, and neither line on this chart is a real vessel.
- **Reversible kinetics are clipped, and the clipping is announced.** Past equilibrium the engine's rate goes negative and those points are excluded from the area, with the exclusion stated in the chart annotation. An integral run through them would be arithmetic, not physics.
- **The stirred tank's volume is not a knob here**, for a recorded reason: the witness declares it inline, out of reach of the override grammar the classroom knobs use. It always runs its authored value. This is stated in the tool's own provenance line rather than left to be discovered.

Where the derivation lives

Derivation. Theory Guide, *Plug-flow reactor* (`ch:pfr`) — the differential balance, the RK4 integration, and the first-order analytic check that gives the canonical comparison at $k\tau = 1$ ($X^{\text{PFR}} = 1 - e^{-1} = 0.632$ against $X^{\text{CSTR}} = 1/(1 + k\tau) = 0.500$), which is this construction in closed form for one special case. The stirred tank's own extent formulation is in *Continuous stirred-tank reactor* (`ch:cstr`).

3.6 Multiple steady states: the Van Heerden diagram

Declared anchor: `ch:cstr`.

The question

The same reactor, the same feed, the same everything — so why does it sometimes sit cold and sometimes run hot, and could it be made to sit in between?

What to vary, what to watch

The diagram is heat **generated** $G(T)$ against heat **removed** $R(T)$, both drawn from the engine's own scan of the reactor's steady-state equation. Generation is a sigmoid: the rate rises exponentially with temperature and then saturates once the reactant is gone. Removal is a straight line, because in an adiabatic vessel the only sink is the sensible heat that leaves with the product. A straight line can cut a sigmoid three times.

Try it now. Drag `T_guess` slowly from 280 K to 430 K and watch the *reported* marker. For a long stretch nothing happens; then it jumps to another crossing on a diagram that has not moved at all. Same equipment, same feed, same equations — a different answer, decided by where you started. That is not a numerical artefact; it is why real reactors have a start-up procedure.

- **`T_guess`** — the decisive knob. The engine finds *all* the roots, prints them, and reports the one nearest your guess.
- **Feed temperature** — slides the removal line. Push it up far enough and the two lower crossings merge and vanish; the reactor can then only be ignited.
- **Reactor volume** — stretches the generation sigmoid against a removal line that does not move with it.
- **The `steadyStates` count and the ignition span** — how many crossings there are, and how far the reactor jumps between the coldest and the hottest.

The misconception the construction exposes

Common pitfall. **That three answers means the solver is broken.** Students trained on "the equation has *an* answer" read multiplicity as a defect — a bad initial guess, a tolerance too loose, a bug to be reported. It is the physics of the system, and the diagram is the proof: an exponential generation curve and a linear removal line intersect three times over a range of conditions, and each intersection is a genuine steady state of the same hardware. The initial guess is not a numerical convenience in this problem. It is the start-up procedure, written as a number.

The one that is harder, and better:

- **That the middle state is merely difficult to converge to.** It is not difficult; it is *unstable.* A steady-state solver will sit on it happily, which is exactly what makes it dangerous pedagogy: the number exists and the reactor cannot hold it. Nudge the temperature up and generation exceeds removal, so it climbs further; nudge it down and it falls to the extinguished state. "The solver converged" and "the plant can run there" are different claims, and this is the cleanest demonstration in the corpus that the second does not follow from the first.
- **That the ignited state is the one you want.** It is the one with the conversion. It may also be past a selectivity limit, past a catalyst's sintering temperature, or past the vessel's design pressure — none of which appear on a diagram whose only axes are temperature and heat.

What the picture does not say

- **It is a steady-state diagram.** It says which states *exist.* It does not say how the reactor travels between them, how long ignition takes, whether it overshoots, or what the transient does to the catalyst. That is the dynamic simulator's subject.
- **The instability flag is an ordering, not an analysis.** With exactly three roots the tool marks the middle one, which is arithmetic on the engine's own root list. With any other count it makes *no* stability claim at all — a restraint worth pointing out to a class, because it is the difference between a result and a habit.
- **The removal line is straight because this reactor is adiabatic.** Add a cooling jacket at a fixed coolant temperature and it is still straight, with a different slope and intercept; let the coolant temperature vary along the vessel and it is not straight at all, and the geometry of "how many crossings" has to be re-argued.
- **The kinetics are a declared teaching variant.** The witness's own header says so, and its Arrhenius parameters are deliberately *not* knobs: a construction about start-up should not silently become a construction about a different chemistry.

Where the derivation lives

Derivation. Theory Guide, *Continuous stirred-tank reactor* (`ch:cstr`) — the material balance in the reaction extent and the one-dimensional solve the engine performs, which is the equation whose roots this diagram displays. The witness case `tutorials/steady/reactors/cstr05_multiplicity` carries the generation-versus-removal argument in its own `flowsheetDict` header, and is the fullest written statement of the construction in the tree; the tool quotes it.

4 The correlation library, and how to read a disagreement

Every construction in §3 rests on correlations, and a simulator normally hides them. It picks one, computes with it, and prints a number. The student learns that the friction factor *is* 0.0223, in the same tone of voice the simulator uses for the molar mass of water.

It is not. It is one correlation's answer, and which one was chosen is part of the answer.

This section is about the instruments that make that visible. They are not charts — they run under `choupoProps` and print — but they teach something no chart can, because the thing to be seen is a *disagreement between authorities*, and a chart of one curve cannot show it.

4.1 What a correlation carries

A correlation in Choupo is an object, and it is required to answer four questions about itself:

- **What am I?** — the expression, evaluated.
- **Where do I stop being valid?** — the window its own authors stated, in words, printed on every run.
- **Who says so?** — a dated primary citation. A correlation whose source lives in a code comment is one the reader cannot check.
- **Can I still prove it?** — a *verify* against a published anchor, re-run every time, with the deviation printed beside the source.

The fourth is the unusual one. A self-check that nobody wires up is a promise, not a check; running it on every use is what turns it into one.

4.2 The friction factor: four answers to one question

Declared anchor: `ch:hydraulics-pressure`.

The question

Water at $\mathrm{Re} = 10^5$ *in a commercial steel line,* $\varepsilon/D = 10^{-3}$*. What is the Darcy friction factor?*

Run `tutorials/props/hydraulics/moody01_friction_correlations` and four correlations answer:

correlation	f	
Blasius	0.017770	**outside its window**
Churchill	0.022343	
Colebrook	0.022175	
Haaland	0.021966	
spread over all four		25.74 %
spread over the three inside their windows		1.72 %

The two numbers, and why there are two

Read the last two rows together, because either one alone is misleading.

The correlations disagree by 26 % is what the first row says, and it is false. Three of them agree within 1.7 % — which for a correlated quantity is agreement, not disagreement.

What produces the 25.74 % is Blasius, and Blasius is not wrong. It is being *misused.* Blasius carries no roughness term at all: it was fitted to smooth pipes and knows nothing about a wall. Asked about $\varepsilon/D = 10^{-3}$ it answers anyway, confidently, twenty per cent low, and there is nothing about the number 0.017770 that looks wrong.

That is the whole content of a validity window, and it is why this run prints one for every correlation on every call. The window is not bureaucracy attached to the formula. It is the only thing standing between a plausible number and a reader.

Try it now. Set `relRough` to 0 — a hydraulically smooth pipe — and run again. Blasius comes back inside its window, the two spreads collapse onto each other, and the four answers agree to a few per cent. Now raise `relRough` to 10^{-2} and watch the gap between the two spreads widen: the more the wall matters, the more expensive Blasius's silence about it becomes.
Then set `Re` to 1500. Three of the four announce that they have substituted the exact laminar law $f = 64/\mathrm{Re}$,

because they are turbulent fits and say so. Churchill does not need to: its single expression contains the laminar law as its own limit, which is why it is the default and why its self-check is anchored there.

What the anchors actually test — and it differs

The run prints a deviation for each correlation against a published anchor. It would be easy to read four small numbers as four validations. They are not the same kind of statement, and the output says which is which:

- **Blasius** is checked against its own closed form. This tests the *arithmetic* and nothing else — and the run says so, so that a deviation of 0.000 % cannot be mistaken for agreement with a measurement.
- **Colebrook** is checked in the fully-rough limit, where its implicit equation collapses to a closed form. This tests that the fixed-point iteration *arrives*.
- **Haaland** is checked against *Colebrook*, because agreement within about 2 % is precisely the claim Haaland published. Anchoring him to anything else would be checking a claim nobody made.
- **Churchill** is checked against the exact laminar law. This is the strongest of the four: it is a property the expression was built to have and that no coding error preserves by accident.

What the picture does not say

None of these four is checked against experimental data. Choupo carries no friction measurements at all. Two of the anchors are self-consistency by construction and one is agreement with another correlation; only Colebrook's tests an algorithm against a closed form the equation itself provides. The run passes, and what it establishes is that the code computes what the papers wrote — not that the papers are right.

The bench also never says which correlation to use. That depends on the pipe: its roughness, whether the flow is developed, whether the duct is even circular. A tool that ranked them would be making the engineer's decision quietly, which is exactly what a silent default does.

And nothing here speaks about the transition band, $2300 < \mathrm{Re} < 4000$, where no correlation was regressed and the flow itself is not single-valued. Correlations evaluated there are marked outside their window, and that mark is the entire answer available.

4.3 One family so far

This discipline covers the convective heat-transfer correlations (`heatTransferBench`) and, since 2026-08-25, the friction factor (`frictionBench`). It does *not* yet cover the rest: the engine computes Wilke–Chang, Chilton–Colburn, Thiele, Weisz, Joback, Lee–Kesler, Rackett and Watson, and those remain internal to the operations that use them — correct, cited in comments, and invisible to a student.

Saying so is the point. A guide that implied an exhaustive catalogue where two families exist would be committing the error this whole section is about: presenting one answer as though it were the only one.

5 One number for "like dissolves like"

A student asked to pick a solvent has a shelf of bottles and no way to rank them. The oldest useful answer is a single number per liquid, and it is worth teaching precisely because it is *almost* right — the places where it fails are as instructive as the places where it works.

5.1 Where the number comes from

Vaporising a liquid pulls its molecules apart. The energy that costs, per unit volume, is the *cohesive energy density*; its square root is Hildebrand's solubility parameter:

$$\delta \;=\; \sqrt{\frac{\Delta H_{\mathrm{vap}}(T) - RT}{V_m}} \tag{1}$$

Two things in that expression are worth stopping on.

The RT is the work the vapour does pushing the atmosphere aside. Subtracting it turns an *enthalpy* of vaporisation into an *energy* of vaporisation, and only the second one measures cohesion. Leaving it in gives a δ a few per cent high — small enough to look right, which is what makes it a favourite error.

And δ is a function of temperature that nobody writes the temperature next to. As a liquid approaches its critical point the cohesive energy goes to zero, and so does δ: there is no longer a liquid to pull apart. Published tables are at 25 °C; a δ quoted without a temperature is a number from an unstated state.

5.2 Running it

`choupoProps` with a `solubilityParameter` operation.

Choupo *derives* δ — it is never stored. Every input is already in the component record (the latent heat at T_b, the critical temperature, the liquid molar volume), so a stored δ would be a second home for a fact the record already fixes, free to drift from it. The latent heat at T comes from the same Watson correlation the enthalpy legs use, not a second copy of it.

The output has three parts: what each substance's δ is and what it was computed from; how far that sits from an independently published value where the record carries one; and the pairwise table $|\delta_i - \delta_j|$, which is what the question actually was.

5.3 The column that makes it checkable

Try it now. Look at the *published* column before you look at anything else.
Equation 1 has three chances to go quietly wrong: a unit slip in V_m, the missing RT, and reporting $\mathrm{Pa}^{0.5}$ where the literature uses $\mathrm{MPa}^{0.5}$ — a factor of 31.6 that still looks like a solubility parameter. None of the three raises an error.
What catches them is that some records carry a δ somebody else computed, independently, and the run prints the deviation. Across the substances where both exist the derivation reproduces the published value to well under a per cent; a deviation of several per cent is therefore a *finding* and not noise.
This is the same device as the Z_c column in a property data bank: a redundant determination turns a plausible wrong number into a visible disagreement. A tool that computes something you cannot check is a tool that is right until the day it is not.

5.4 Where one number stops

Hildebrand carries no hydrogen bonding. Ethanol ($\delta \approx 26\ \mathrm{MPa}^{1/2}$) and nitromethane ($\approx 25.1$) sit almost on top of each other, and a formulator knows they behave differently — one donates hydrogen bonds and the other does not. The theory cannot express the difference, because it compressed every kind of intermolecular force into one scalar.

That is why the coatings industry uses Hansen's three-way split: dispersion, polar and hydrogen-bonding contributions, with miscibility judged by a distance in that space rather than on a line. Choupo does *not* implement it, and the run says so every time: Hansen's parameters are a separate dataset this project does not hold.

So the honest reading of a close pair is *candidate*, never *verdict* — and the run prints exactly that sentence, on every call, rather than letting a small number imply more than it carries.

Try it now. Put water in the list. Its δ is enormous (≈ 48 MPa$^{1/2}$), far from every organic on the shelf, and the pairwise table will duly report that water mixes with nothing. Water and ethanol are of course miscible in all proportions.
That single row is the clearest lesson the tool gives: the number is a measure of *how much* energy holds a liquid together, and says nothing about *what kind*.

6 Constructions not yet documented here

The tools written up above are not the whole registry: several live tools are *not* written up in this guide. The six below are listed with what exists and what is missing, because an admitted gap is a better guide than six thin sections; the remaining ones — the theory lessons reached from the Methods workspace, and the constructions added since this section was written — carry their lesson text in the app, and the registry (`gui/src/ui/methods/registry.ts`) is the authority on the complete list. Each is a working tool: what is absent here is the lesson material, not the construction.

For every entry below, what is missing is the same three things:

- the **misconception** the construction exposes — the part that has to be observed in a classroom before it can be written down, and the part that cannot be guessed from the source;
- **what the picture does not say** — the limits, stated with the same care as the six above;
- a **worked "what to vary, what to watch"** sequence, tried on the tool rather than inferred from its knobs.

Psychrometric chart (`tool=psychro`)

The humid-gas state map: saturation, relative-humidity lines, adiabatic-saturation and true wet-bulb lines, over any carrier gas and condensable pair the catalogue can serve. Every line is engine-computed. *Declared anchor:* `ch:drying`.

Drying curve, batch tray (`tool=drying`)

The two classical drying plots side by side on one run — moisture against time, and rate against moisture — with the critical moisture visible as the corner where a flat rate begins to fall. The tool is careful about one thing that deserves writing up properly: the engine omits the break time entirely when the crossing was never observed, and the tool reports *which* absence it is rather than drawing a zero. *Declared anchor:* `ch:drying`.

Cooling tower, Merkel (`tool=merkel`)

The saturated-air enthalpy curve above, the operating line below, and the gap between them shaded — that gap is the driving force the packing has to buy. A chip holds the four-point Chebyshev hand method against the engine's converged integral. *Declared anchor:* `ch:coolingTower`.

Batch still, Rayleigh (`tool=rayleigh`)

The graphical Rayleigh integration: the area under $1/(y^* - x)$ between the pot's final composition and the charge *is* $\ln(W_0/W)$, drawn over the engine's own equilibrium curve and judged against the engine's rigorously integrated still. *Declared anchor:* `ch:rayleigh`.

Pump against system curve (`tool=pump-system`)

The operating point as an intersection: the pump's pressure rise falling with flow against the pipe system's demand rising with it. The tool is explicit that the "pump curve" is whatever the pump *model* published, and greys it out with the caveat spelled in full when the model was specified by a fixed pressure rise and the curve is therefore the specification echoed back. *Declared anchor:* `ch:rotating`.

Adsorption breakthrough (`tool=breakthrough`)

The S-shaped breakthrough curve against the ideal square wave drawn at the engine's stoichiometric time: the mass-transfer zone consumes bed capacity long before the bed is saturated. *Declared anchor:* `ch:adsorption`.

7 Using a tool in a lesson

Three patterns that work, drawn from how the tools are built rather than from how they could be used.

Turn one knob, and predict first

Every classroom tool re-solves in the browser on each knob change, which makes the interesting move cheap: ask for a prediction, then turn the knob. The productive knobs are the ones whose consequence is counter-intuitive — $\Delta T_{\min}$ on the pinch tool, area on the ε–NTU chart, `T_guess` on the Van Heerden diagram, the solute loading on the Kremser tool. All four are documented above with what should happen.

Make the deviation chip the discussion

Several tools put a classical hand method beside the engine's answer and report the gap. That gap is the lesson, and its size is not the point — its *cause* is. On the Kremser tool the gap is the isothermal assumption failing, and it grows the more exothermic the duty. On the cooling-tower tool the gap is nearly zero, and saying so plainly rather than manufacturing drama is itself worth modelling for students: a hand method that reproduces a converged integral to five figures has been *validated*, and that is a result.

Read the refusal

An engine that refuses is teaching. A pinched McCabe–Thiele construction, an infeasible cooling tower, an absorber whose stripper twin does not activate the tool, a drying run that never reached its critical moisture — each of those is a message with a reason in it. The tools show the engine's own words verbatim rather than a generic failure, and the reason is usually the physics of the case rather than a mistake in the input.

> **Key idea.** All models are wrong; some are useful. Everything in this plane is built so a student can see *which* assumption is doing the work — and what happens on the day it stops being true.

8 Notes for maintainers

One anchor, and it must resolve

Each tool's registry entry names a Theory Guide destination. The preamble shared by all the manuals sets `destlabel=true`, so every `\label` in the Theory Guide becomes a PDF named destination and the book icon jumps straight to it. An anchor the guide does not define fails *silently* in a browser — the PDF simply opens at page one — so a test in the application checks every declared anchor against the guide source. A tool without a resolvable anchor is not finished.

Anchors are chapter-level, and some chapters are broader than their tool

The declared anchor points at the section that owns the tool's *unit operation*, which is not always the section that derives the tool's *construction*. Where the two differ, the *Where the derivation lives* paragraphs above name both destinations rather than pretending the first covers everything. Two cases are worth knowing about before adding a seventh tool section:

- A construction may be the hand method for a chapter that derives the *rigorous* model instead — the staircase against the stagewise column, the Levenspiel areas against the differential balance.
- A construction may be argued most fully in its *witness case's* own dictionary header rather than in the manual. That is a real source and this guide cites it as one, but it is not a substitute for a derivation, and a section that has to reach for it is flagging a gap in the Theory Guide rather than filling one.

When a tool changes

A knob added or removed, a chip relabelled, a witness re-parameterised: each of those can invalidate a *What to vary, what to watch* block here, and none of them will fail a build. The blocks are written to name knobs by their on-screen labels so a mismatch is at least visible to a reader with the tool open. Prose staleness is deliberately not gated in this project — a text search cannot tell a live claim from a recorded one — so the check is human, and it belongs in the same change that touches the tool.